Robust Cross-Modal Remote Sensing Image Retrieval via Maximal Correlation Augmentation

Zhuoyue Wang, Xueqian Wang, *Member, IEEE*, Gang Li, *Senior Member, IEEE*, and Chengxi Li

***Abstract*— Most of the existing studies regarding cross-modal content-based remote sensing image retrieval (CM-CBRSIR) focus on reducing/enlarging the Euclidean distances of cross-modal (CM) data with the same/different content in a common feature space. The advantages of using Euclidean distance lie in its straightforwardness. However, the Euclidean distances of CM data features are sensitive to the outlier data and may lead to non-robust retrieval performance, particularly in the case of noisy images with low quality. To address this issue, we propose a robust Hirschfeld–Gebelein–Rényi maximal correlation (HGRMC) augmented algorithm for CM-CBRSIR in this work, named by HGRMC augmented CM-CBRSIR (HAC). In HAC, not only the projected features of CM data in Euclidean distance space but also maximal correlation information of HGRMC are learned during the training phase of the retrieval model, where HGRMC is additionally used to capture the statistical dependency between CM data to enhance the retrieval performance with the strongly noisy input data. In the retrieval phase, we also develop a fusion scheme based on the Dempster–Shafer (DS) evidence theory to combine the superiorities of Euclidean distance and HGRMC correlation criteria. Extensive experimental results demonstrate that our proposed HAC algorithm provides better and more robust retrieval performance in comparison with existing state-of-the-art CM-CBRSIR methods.**

***Index Terms*— Cross-modal image retrieval, Dempster–Shafer (DS) evidence theory, Hirschfeld–Gebelein–Rényi maximal correlation (HGRMC), remote sensing data management.**

I. Introduction

NUMEROUS Earth observation sensors generate massive amounts of remote sensing data every day. With the explosive growth of remote sensing data, it is crucial to investigate effective content-based remote sensing image retrieval (CBRSIR) methods to identify relevant images quickly and easily from large data archives [2], [3]. In recent years, CBRSIR methods using unified-modal (UM) images have been well developed [4], [5], [6], [7], [8], [9], [10], [11]. For instance, Liu et al. [6] develop a feature learning method for UM-CBRSIR, which employs an adversarial hash learning model to generate compact hash codes from multiscale features. Sun et al. [11] propose an unsupervised deep hashing method, which leverages soft pseudo-labels and local similarity matrices to capture intercluster and intracluster similarities. However, the requirement of UM-CBRSIR that the modality of samples to be retrieved is necessarily the same as that of data archives reduces the flexibility of retrieval models.

Cross-modal CBRSIR (CM-CBRSIR) facilitates the capability of image retrieval models to locate and prioritize relevant data across disparate modalities [12], [13], [14], [15], which has a diverse range of potential applications. For instance, it can enhance disaster response and management by facilitating the rapid retrieval of relevant imagery in disaster search and rescue tasks [16]. Moreover, it plays an essential role for meteorological analysis [17]. It also critically supports forest monitoring [18], contributing to conservation and resource management.

In this article, we mainly focus on the problem of CM-CBRSIR using optical and synthetic aperture radar (SAR) images, such as [19] and [20]. In practice, optical images with good lighting conditions are high-resolution and are easy to understand by humans, and SAR offers dependable images irrespective of atmospheric or lighting variations. SAR and optical images together form a comprehensive basis for remote sensing analysis. Note that a key issue of CM-CBRSIR is to bridge the heterogeneity gap caused by the difference of image modalities. Existing CM-CBRSIR methods can be broadly categorized into CM image hashing [19], [20], [21], [22], [23] and CM image retrieval with real-valued features [13], [15], [24], [25].

CM hashing is an efficient strategy, which aims to reduce storage requirements and retrieval times in large-scale CM image retrieval, where real-valued features are transformed into compact binary codes. Traditional handcrafted methods, such as cross-view hashing (CVH) [26] and collective matrix factorization hashing (CMFH) [27], have shown potential in addressing the heterogeneity gap between different modalities. However, these methods [26], [27] show limited performance gain when handling CM data with nonlinear subspace structures. Due to the superior capability of deep neural networks (DNNs) to model intricate relationships between different modalities, there is a surge in the use of DNNs in CM hashing methods. Unsupervised deep CM hashing methods

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use unlabeled data to generate hashing codes [21], [28], [29]. A common strategy in unsupervised CM hashing is contrastive learning [21], [28], [30], which better leverages the ranking structure of data to optimize hash codes for improved retrieval performance. For deep supervised hashing methods, Jiang and Li [31] propose a deep cross-modal hashing (DCMH) method, which uses DNNs to learn unified binary codes for different modalities in an end-to-end fashion via likelihood maximization. In [22], a source-invariant deep hashing convolutional neural network (SIDHCNN) is introduced, adeptly learning binary hash codes through the integration of two distinct subnetworks tailored to the spatial–spectral resolutions specific to different remote sensing image modalities. Sun et al. [20] propose a multisensor fusion and semantic preservation-based deep hashing method between optical images and SAR images, where synthetically generated images are utilized to augment the retrieval model. Guo and Guan [32] focus on the CM vessel image retrieval task, where a deep cascaded network with a self-attention mechanism is used to extract features of low-resolution vessel images. A center-based method [23] is proposed for CM-CBRSIR, in which consistency centers are utilized to reduce discrepancies across multiple image sources.

In contrast to CM hashing methods, real-valued representation-based retrieval methods retain continuous values in the feature representation space, which helps to preserve more information from the original data in comparison with hashing codes. In practice, real-valued representation-based retrieval methods often yield higher retrieval accuracy than hashing code-based methods [33] with slightly higher requirements for computation and storage sources. Canonical correlation analysis (CCA) [34] and generalized multiview analysis (GMA) [35] methods are two classical real-valued representation-based retrieval methods but only based on limited image features. For deep unsupervised real-valued methods, deep CCA [36] extends the CCA-based models by integrating the DNNs with powerful feature extraction capabilities. In [37] and [38], autoencoders are introduced to directly minimize the difference between the generated features from different modalities but ignore the similarity ranking. Contrastive learning has been applied to model the ranking relationship based on the relevance of samples, where distance-based triplet loss is used to preserve the ranking structure to queries [39], [40]. For deep supervised real-valued methods, Xiong et al. [41] employ generative adversarial networks (GANs) for image style transformation, facilitating retrieval among images within the same modality after the transformation. In [42], adversarial learning is introduced, where the adversarial process between the feature projector and the classifier drives the network to generate more robust and modality-invariant representations. To ensure the learned representations are consistent regardless of their originating modality, in [24] and [43], both the discrimination loss and the Euclidean distance-based intermodal invariance loss are formulated, which contribute to learning more discriminative features, thereby achieving enhanced retrieval accuracy. In [15], an asymmetric multimodal feature matching network is proposed to tackle the issues of multiscale scarcity and target redundancy during retrieval. Chaudhuri et al. [44] propose a zero-shot training protocol, aiming to address the challenges of minimal labeled data and unobserved classes in CM-CBRSIR. Among these deep supervised methods, notable advancements have been made, primarily through novel design in network architectures for enhanced feature extraction [15], [41], [44], as well as through the development of novel learning paradigms after feature extraction [24], [42], [43]. However, the robustness of the learning metric under suboptimal data conditions, such as the presence of noise, is often overlooked. In contrast, our proposed method focuses on robust learning, even under nonideal conditions, to improve the reliability of CM-CBRSIR.

Most existing DNN-based real-valued methods [42], [43], [45], [46] rely on Euclidean distance (i.e., L_2-norm distance) as the primary constraint during training and perform direct comparisons of CM features and ranking of the retrieval results during the retrieval phase. The utilization of Euclidean distance is a well-accepted practice owing to its simplicity and effectiveness in measuring geometric properties of the data, and the advantage of not requiring additional statistical estimation makes it computationally efficient. However, image noise, which is a pervasive and significant challenge in the context of Euclidean distance-based CM-CBRSIR DNN methods [47], [48], is widely present in remote sensing images, e.g., speckle noise in SAR images [49] and Gaussian noise in optical images [50]. Particularly in CM-CBRSIR systems, image noise not only distorts the intrinsic information of images but also impairs the correspondence relationship between images from distinct modalities during the training process, which increases the difficulty of retrieval. However, the majority of related methods do not specifically account for scenarios with significant noise. While using the Euclidean distance is computationally efficient and straightforward in its geometric interpretation of the data, it is particularly vulnerable to the distortions caused by image noise. In situations where image noise alters the feature entries, the Euclidean distance can be misled, causing incorrect rankings in the retrieval results [51], [52], [53]. This motivates us to investigate better model of the relationship between different modalities to boost the performance of CM-CBRSIR.

In this article, we introduce the Hirschfeld–Gebelein–Rényi maximal correlation (HGRMC) [54], [55], [56] criterion to generate an improved DNN-based CM-CBRSIR model. As the nonlinear extension of Pearson's correlation, HGRMC is a measure of nonlinear statistical dependency, which captures a set of orthogonal features with the maximum amount of correlation information of two variables. HGRMC has been demonstrated to be valuable in the design of a variety of machine learning algorithms, such as feature selection and classification [57], [58]. In comparison with Euclidean distance criterion, HGRMC provides a more robust framework for uncovering underlying statistical dependencies between variables, even amidst noise-induced distortions. By maximizing correlation through nonlinear transformations under orthogonality constraints, HGRMC is designed to reveal stable relationships between variables, thereby offering an increment in terms of robustness in noisy scenarios.

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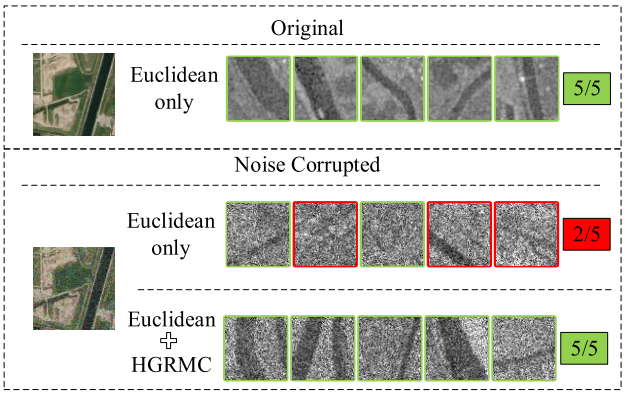


Fig. 1. Query examples with the corresponding top-5 retrieved images under both original and noisy scenarios based on only Euclidean features and the fusion of Euclidean and HGRMC features, where the samples in green rectangles were correct returned images, and samples in red rectangles were false positives. The top correctly retrieved images may vary, as images within the same class were not ranked by priority.

Fig. 1 illustrates an example of the retrieval performance of models utilizing Euclidean features alone and those integrating HGRMC features, under both normal and noisy scenarios. Fig. 1 shows the top-5 retrieved images for a given query. In the normal scenario with the low noise level, we retrieve all correct images by using Euclidean feature solely. However, the scenario changes under noisy conditions, where the Euclidean-based model yields three incorrect returns, indicating a susceptibility to noise-induced distortions. Conversely, the model integrating HGRMC features demonstrates a notable resilience to noise, accurately retrieving all the images even in the noisy environment.

In our study, we propose a novel CM-CBRSIR algorithm, HGRMC augmented CM-CBRSIR (HAC), to exploit effectiveness of HGRMC in the context of CM-CBRSIR. The main contributions of our HAC are twofold.

1) During the training phase, HAC jointly learns features under the Euclidean distance constraints and HGRMC constraints. Thanks to the capability of HGRMC to capture underlying statistical dependencies amidst noise, HGRMC in retrieval model learning improves the robustness of retrieval performance compared with the existing Euclidean distance-based CM-CBRSIR methods, particularly in noisy scenarios.
2) During the test phase, we design a fusion algorithm to combine Euclidean and HGRMC criterions based on the Dempster–Shafer (DS) evidence theory to provide the final retrieval result. The DS-based fusion scheme excels in resolving conflicting evidence in noisy scenarios and provides robust retrieval performance.

Experimental results underscore the superior performance of our proposed method in comparison with other state-of-the-art methods, particularly in strongly noisy environments.

The rest of this article is organized as follows. Section II presents a review of HGRMC and the related works. In Section III, we introduce preliminary knowledge that is crucial for understanding the proposed method. Section IV provides a detailed description of the proposed HAC algorithm. In Section V, we conduct a performance analysis of HGRMC and Euclidean features to illustrate the necessity for feature fusion in enhancing retrieval robustness. Section VI provides in-depth experimental comparisons between the proposed HAC approach and existing methods. Finally, Section VII concludes this article with several remarks at plausible research points in the future.

II. Related Work

In this section, we introduce related work on HGRMC and DS evidence theory, alongside their applications in retrieval tasks.

HGRMC has been recognized for its capability to capture nonlinear dependency and has been the foundation for various applications. Hirschfeld [54], Gebelein [55], and Rényi [56] give definitions and provide the basic concepts of HGRMC. Recently, a comprehensive introduction by Huang et al. [57] traces the historical evolution and introduce various properties of the HGRMC. In [59], the sample complexity of HGRMC is investigated to provide rigorous insights into its scalability and robustness in the context of large datasets, which is widely used in machine learning. In [60] and [61], the computational complexity problem of HGRMC is discussed and scalable methods to estimate HGRMC are proposed.

The versatility of HGRMC has led to its adoption in various applications, ranging from theoretical interpretation of DNN [62], [63], [64] to various important machine learning tasks [65], [66], [67], [68], [69]. Huang et al. [63] delve into an information-theoretic approach to design functional representations that extract the hidden common structure shared by a set of random variables. Xu et al. [62] provide an information-theoretic interpretation to the feature extraction process in DNNs. HGRMC is also used to demonstrate the information-theoretic optimality of distributed learning [64]. Expanding on the foundational concepts of HGRMC, Lee et al. [66] explore the use of HGRMC in the context of multisource transfer learning, which leverages the principles of HGRMC to combine multiple pretrained neural networks for few-shot learning of classifiers for previously unseen classes. In the field of algorithmic fairness in machine learning, an information-theoretic perspective is adopted in [65], where HGRMC is used as a constraint for expressing fairness constraints to enforce both independence and separation-based fairness criteria. HGRMC has also been employed in the domain of person recognition with empirical results underscoring its robustness against noise [67].

In [69], HGRMC is employed in the training process for CM retrieval, but it does not consider the valuable information encapsulated in the Euclidean distance, which is a powerful tool to mitigate overfitting problems when training data are rare. Our approach integrates both Euclidean distance and HGRMC criteria, not only during the training phase but also in the retrieval phase, thereby exploiting the complementary strengths of both metrics to improve the robustness and accuracy of CM-CBRSIR.

The DS evidence theory, initially introduced by Dempster [70] and later expanded by Shafer [71], presents a flexible and general approach to represent both imprecision and uncertainty, which has various applications in information fusion from different sources. In [72], the DS evidence theory is applied to multisource remote sensing classification task

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in an unsupervised manner by integrating data from multiple sensors to enhance the classification accuracy. Han et al. [73] propose a novel trusted multiview classification based on the DS evidence theory, which produces trusted classification decision on multiview data. In addition, the DS evidence theory is applied within a complex network model to address the issues of conflicting evidence and high computational complexity from a network graph perspective [74]. In our work, we leverage the DS evidence theory to combine the information learned from Euclidean distance and HGRMC to generate reliable ranking results in the retrieval phase. Compared with our earlier published version [1], we delve deeper into the rationale behind the model design and conduct further analysis and experiments to investigate the robustness of our proposed model under nonideal data conditions.

III. Preliminary Knowledge

This section provides an introduction of HGRMC used in our proposed HAC method.

HGRMC [54], [55], [56], [57] serves as an advanced statistical tool, effectively quantifying the degree of nonlinear dependence between random variables.

Definition 1 (HGRMC): For two jointly distributed random variables with range $\mathcal{X}$ and $\mathcal{Y}$:$X \in \mathcal{X}$, $Y \in \mathcal{Y}$, HGRMC with k features is defined by

$$\left(\mathbf{f}^*, \mathbf{g}^*\right) = \underset{\substack{\mathbf{f}:\, X\to\mathbb{R}^k,\ \mathbf{g}:\, Y\to\mathbb{R}^k \\ \mathbb{E}[\mathbf{f}(X)]=\mathbb{E}[\mathbf{g}(Y)]=\mathbf{0} \\ \mathrm{cov}[\mathbf{f}(X)]=\mathbf{I} \\ \mathrm{cov}[\mathbf{g}(Y)]=\mathbf{I}}}{\operatorname{argmax}} \mathbb{E}\left[\mathbf{f}^{\mathrm{T}}(X)\mathbf{g}(Y)\right], \tag{1}$$

where $\mathbf{f}^* = (f_1^*, \ldots, f_k^*)^{\mathrm{T}}$ and $\mathbf{g}^* = (g_1^*, \ldots, g_k^*)^{\mathrm{T}}$ denote the maximal correlation functions, $\mathrm{cov}(\cdot)$ denotes the covariance matrix, $\mathbf{I}$ is the identity matrix, and $\mathbb{E}(\cdot)$ represents the expectation.

In context of CM-CBRSIR, X and Y represent data from different modalities, and $\mathbf{f}$ and $\mathbf{g}$ are their corresponding nonlinear transformation functions. In (1), the maximal correlations of CM data are captured based on $\mathbf{f}$ and $\mathbf{g}$, which nonlinearly project modality-specific data into a shared latent space. Zero-mean constraint $\mathbb{E}[\mathbf{f}(X)] = \mathbb{E}[\mathbf{g}(Y)] = \mathbf{0}$ centers the transformed data in the shared latent space. $\mathrm{cov}[\mathbf{f}(X)] = \mathrm{cov}[\mathbf{g}(Y)] = \mathbf{I}$ in (1) serves as a decorrelation and normalization constraints, reducing redundant features and mitigating the risks of perceived inaccurate dependence relationships caused by data noise. In our HAC method, we employ HGRMC to capture the maximal correlation information between distinct modalities, coupled with its decorrelation mechanism, allowing HGRMC to more effectively and robustly capture underlying statistical dependencies between CM remote sensing data.

Note that the direct calculation of HGRMC in (1) leads to high computational complexity [60], [61]. In order to alleviate the computational burden of HGRMC, we choose to use a variational approximation of the original HGRMC, termed soft-HGR [60]

$$\mathrm{HGR}_s(X, Y) = \max_{\substack{\mathbb{E}[\mathbf{f}(X)]=0 \\ \mathbb{E}[\mathbf{g}(Y)]=0}} \mathbb{E}\left[\mathbf{f}^{\mathrm{T}}(X)\mathbf{g}(Y)\right] - \frac{1}{2}\mathrm{tr}(\mathrm{cov}(\mathbf{f}(X))\mathrm{cov}(\mathbf{g}(Y))), \tag{2}$$

where $\mathrm{tr}(\cdot)$ represents the matrix trace. Soft-HGR does not require time-consuming matrix inversion and decomposition in whitening constraints in (1). It has been theoretically established that the optimal feature transformations of soft-HGR are linearly transformable from the HGRMC, implying that they span equivalent feature spaces [60].

IV. Proposed HAC Method

Assuming that we have a dataset of triplets $\mathcal{D} = \{(v_i, s_i, y_i)|i = 1, 2, \ldots, n\}$, where v_i and s_i represent the ith optical image and SAR image, respectively, and n denotes the size of the dataset, $\mathbf{y}_i = [y_{i1}, \ldots, y_{ic}] \in \mathbb{R}^c$ is the binary label vector, c is the number of image content classes, $\sum_{j=1}^{c} y_{ij} = 1$, and if the ith data pair belongs to the class j, then we have $y_{ij} = 1$. The optical and SAR datasets are denoted by $\mathcal{V} = \{v_i|i = 1, 2, \ldots, n\}$ and $\mathcal{S} = \{s_i|i = 1, 2, \ldots, n\}$, respectively. For simplicity of description, we take the task of searching SAR images in $\mathcal{S}$ when given an optical image query as an example to elaborate our HAC method, where HAC is to accurately extract semantically relevant samples from the SAR dataset $\mathcal{S} = \{s_i|i = 1, 2, \ldots, n\}$.

A. Overview of the Training Stage

The diagram of the training stage of our HAC is shown in Fig. 2. First, we employ convolutional neural networks (CNNs) to separately extract high-level features of optical and SAR images. Next, for each modality, we feed the high-level features into two fully connected networks to obtain a set of features for Euclidean distance optimization (Euclidean features), denoted by $\mathbf{f}_E(v_i)$ and $\mathbf{g}_E(s_i)$ for optical and SAR images, respectively. In addition, we also obtain a set of features for HGRMC optimization (HGRMC features), denoted by $\mathbf{f}_H(v_i)$ and $\mathbf{g}_H(s_i)$ for optical and SAR images, respectively. In order to make full use of the supervised information, we employ a label classifier, denoted as $P(\cdot)$, using Euclidean and HGRMC features to predict the semantic labels of the query image. As illustrated in Fig. 2, we design a loss function combining the semantic prediction loss, the Euclidean distance loss, and the HGRMC loss. The combined loss function is minimized to train the proposed HAC model. The HGRMC loss complements the Euclidean distance loss by enhancing retrieval performance, especially under strongly noisy conditions.

B. High-Level Feature Extraction

The high-level feature extraction module in the proposed HAC method consists of multiple convolutional layers and fully connected layers to learn semantic features from specific modalities. The architecture of convolutional layers for each modality is AlexNet [75]. We select AlexNet for our feature extraction module due to its established efficacy in effectively capturing hierarchical visual features while maintaining computational efficiency, as well as its successful application in CM-CBRSIR studies [12], [20]. In addition, our approach remains adaptable, designed to incorporate other state-of-the-art CNN architectures for feature extraction. We choose the output of fc3 of AlexNet as the high-level modality-specific features. To accelerate the training process, we choose the

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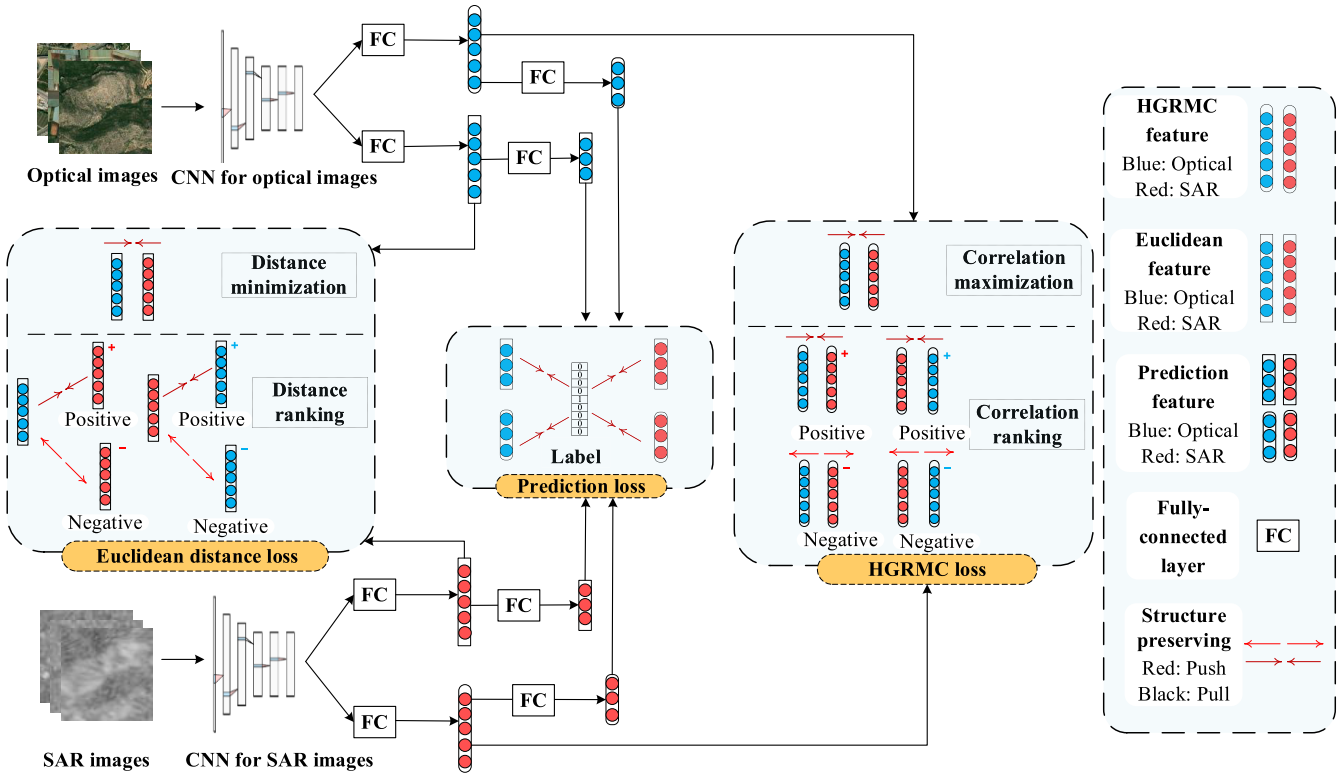


Fig. 2. Training strategy of the proposed method for CM-CBRSIR. First, pretrained CNNs for optical and SAR images are used as the modality-specific feature extraction modules to generate high-level features. For each modality, high-level features are fed into two parallel and one-layer fully connected network to generate Euclidean and HGRMC features. In the end, a classifier is added to generate a prediction feature in order to make use of supervised information.

AlexNet model pretrained on ImageNet [76], which can also provide a robust initialization for training on remote sensing datasets [20], [77]. For each modality, we use two parallel fully connected networks to generate the Euclidean features and the HGRMC features after the high-level feature extraction, where each fully connected network is one layer. Another one-layer fully connected network is employed as the final image content classifier after the Euclidean and HGRMC feature extraction.

C. Loss Function

Our novel loss function integrates the Euclidean distance loss $\mathcal{J}_{\text{EUC}}$, HGRMC loss $\mathcal{J}_{\text{HGR}}$, and prediction loss $\mathcal{J}_P$.

The Euclidean loss consists of the similarity loss $\mathcal{J}_I$ and the ranking loss $\mathcal{J}_R$

$$\mathcal{J}_{\text{EUC}} = \mathcal{J}_I + \mathcal{J}_R, \tag{3}$$

where

$$\mathcal{J}_I = \frac{1}{N}\sum_{i=1}^{N} d(\mathbf{f}_E(v_i), \mathbf{g}_E(s_i)), \tag{4}$$

$$\mathcal{J}_R = \frac{1}{N}\sum_{i=1}^{N} \max\left\{d\left(\mathbf{f}_E(v_i), \mathbf{g}_E\left(s_i^+\right)\right) - d\left(\mathbf{f}_E(v_i), \mathbf{g}_E\left(s_i^-\right)\right) + m, 0\right\} + \max\left\{d\left(\mathbf{f}_E\left(v_i^+\right), \mathbf{g}_E(s_i)\right) - d\left(\mathbf{f}_E\left(v_i^-\right), \mathbf{g}_E(s_i)\right) + m, 0\right\}, \tag{5}$$

where N is the batch size during the training stage; $d(\cdot,\cdot)$ denotes the Euclidean distance; v_i^+ and v_i^- represent positive and negative optical samples for the ith SAR image s_i in the current batch, respectively; s_i^+ and s_i^- represent the positive and negative SAR samples for the ith optical image v_i in the current batch, respectively; and m is the margin parameter to determine the feature discrimination degree between positive and negative samples. A positive sample refers to an image that shares the same class as the anchor image, while the class of a negative sample is different from the anchor image. We choose one positive and one negative sample in the loss function to keep the training process tractable and efficient. The minimization of $\mathcal{J}_I$ in (4) is to minimize the Euclidean distances between high-level features of the training data pairs and the minimization $\mathcal{J}_R$ in (5) is to ensure the correct ranking structure among the data candidates based on their degree of semantic relevance, as quantified by the Euclidean distance.

While the Euclidean distance plays a critical role in the feature learning process thanks to its simplicity and computational efficiency, it is sensitive to noise and outlier points within the data, which can compromise its robustness in image retrieval tasks [51], [52], [53]. The Euclidean distance, without decorrelation and normalization, can misinterpret noisy data, which may rely on redundant information and allow larger magnitude features to dominate when noise distorts these aspects. The direct spatial-based computation of Euclidean distance might interpret these distortions as genuine patterns. HGRMC emphasizes maximizing the correlation between features from distinct modalities that share the same semantic content. The normalization in HGRMC prevents noise from distorting

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feature magnitudes, ensuring consistent scales, while decorrelation promotes orthogonal feature learning and reduces the influence of noise across dimensions. Consequently, HGRMC serves as an effective compensation for the limitations of the Euclidean distance, enhancing the robustness of our HAC model, especially in scenarios with noise and outlier data. Therefore, we introduce the HGRMC loss function into our model to extract the maximal correlation information between SAR and optical modalities. The HGRMC loss is composed of the correlation loss $\mathcal{J}_C$ and the correlation ranking loss $\mathcal{J}_{\mathrm{CR}}$. Minimizing the HGRMC loss is to maximize the pairwise correlation and preserve the ranking structure, i.e.,

$$\mathcal{J}_{\mathrm{HGR}} = \mathcal{J}_C + \mathcal{J}_{\mathrm{CR}}, \tag{6}$$

where

$$\mathcal{J}_C = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{2}\mathrm{tr}\big(\mathrm{cov}\big(\tilde{\mathbf{f}}_H(v_i)\big)\mathrm{cov}\big(\tilde{\mathbf{g}}_H(s_i)\big)\big) - \big[\tilde{\mathbf{f}}_H^{\top}(v_i)\tilde{\mathbf{g}}_H(s_i)\big], \tag{7}$$

$$\tilde{\mathbf{f}}_H(v_i) = \mathbf{f}_H(v_i) - \frac{1}{N}\sum_{i=1}^{N}\mathbf{f}_H(v_i),$$

$$\tilde{\mathbf{g}}_H(s_i) = \mathbf{g}_H(s_i) - \frac{1}{N}\sum_{i=1}^{N}\mathbf{g}_H(s_i), \tag{8}$$

and

$$\mathcal{J}_{\mathrm{CR}} = \frac{1}{N}\sum_{i=1}^{N}\big[\mathcal{J}_C\big(v_i, s_i^{+}\big) + \mathcal{J}_C\big(v_i^{+}, s_i\big) - \mathcal{J}_C\big(v_i, s_i^{-}\big) - \mathcal{J}_C\big(v_i^{-}, s_i\big)\big]. \tag{9}$$

The minimization of the correlation loss $\mathcal{J}_C$ in (7) aims to maximize the HGRMC between the HGRMC features of data pairs from optical and SAR modalities. We achieve this by applying the soft-HGR, as expressed in (2). $\tilde{\mathbf{f}}_H$ and $\tilde{\mathbf{g}}_H$ are the centered transformed features from optical and SAR modalities, respectively. The first term in (7) acts as a soft regularizer, which replaces the whitening constraints of HGRMC to perform the decorrelation mechanism in a more computationally tractable manner. The inner product operation in the second term of (7) is consistent with the objective of the HGRMC for correlation maximization. The correlation ranking loss in (9) is designed to maximize the correlation of HGRMC features from instances belonging to the same category while minimizing the correlation of HGRMC features with distinct semantics.

The last part of the loss function is the prediction loss, which is employed to make use of label information to enhance the retrieval performance. The mean squared error loss is used as the prediction constraint

$$\mathcal{J}_P = \frac{1}{N}\sum_{i=1}^{N}\big[\|P(\mathbf{f}_E(v_i)) - y_i\|^2 + \|P(\mathbf{f}_H(v_i)) - y_i\|^2 + \|P(\mathbf{g}_E(s_i)) - y_i\|^2 + \|P(\mathbf{g}_H(s_i)) - y_i\|^2\big], \tag{10}$$

where a lower value of $\mathcal{J}_P$ indicates a higher precision in label prediction.

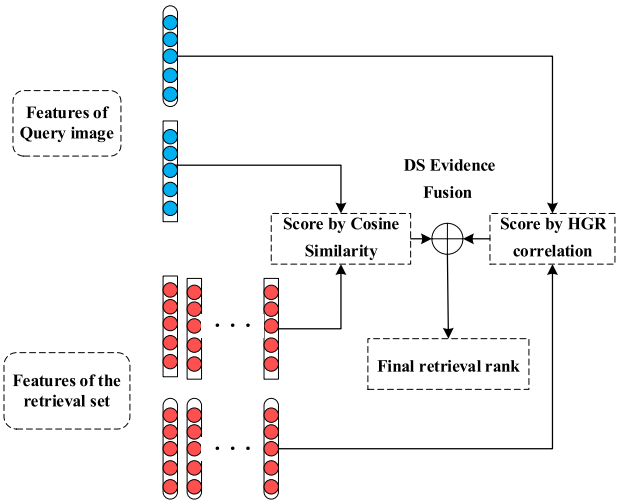


Fig. 3. Retrieval model of the proposed HAC method for CM-CBRSIR.

In summary, we obtain the overall loss function during the training process by integrating the Euclidean loss, HGRMC loss, and prediction loss

$$\mathcal{J} = \mathcal{J}_P + \alpha\mathcal{J}_{\mathrm{EUC}} + \beta\mathcal{J}_{\mathrm{HGR}}, \tag{11}$$

where $\alpha > 0$ and $\beta > 0$ are hyperparameters to control the weights of different loss terms.

D. Retrieval Stage

Once the training process is completed, we search for semantically related samples in the retrieval dataset during the retrieval process, as shown in Fig. 3. First, well-trained CNNs in Section IV-B are used to extract the Euclidean features and HGRMC features of query images and images from the retrieval dataset. Subsequently, we calculate the cosine similarities and HGRMC scores between the features of query images and those in the retrieval dataset, in terms of Euclidean feature space and HGRMC feature space, respectively. The cosine similarities and HGRMC degrees are then normalized to [0, 1] based on their respective minimum and maximum values to obtain the Euclidean score denoted by s_{EUC} and HGRMC score denoted by s_{HGR}.

Next, we combine the Euclidean score s_{EUC} of Euclidean features and HGRMC scores s_{HGR} of HGRMC features. The DS evidence theory, a mathematical framework adept at handling uncertainty and combining evidence from multiple sources, is employed here due to its robustness in managing incomplete or even conflicting information, which is common in noisy scenarios. In this section, we leverage the DS evidence theory to synthesize s_{EUC} and s_{HGR} scores and then obtain the final retrieval decision.

First, we define the hypotheses based on the DS formulation

$$\begin{aligned} &A_1\text{: The score is reliable}\\ &A_2\text{: The score is not reliable}\\ &A_3\text{: The score is reliable or not}\\ &A_4\text{: }\emptyset. \end{aligned} \tag{12}$$

For the Euclidean and HGRMC scores, the mass functions of Euclidean and HGRMC features, i.e., m_1 and m_2, are

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defined as follows:

$$\begin{aligned} m_1(A_1) &= b_1 s_{\mathrm{EUC}}; & m_2(A_1) &= b_2 s_{\mathrm{HGR}}, \\ m_1(A_2) &= b_1(1-s_{\mathrm{EUC}}); & m_2(A_2) &= b_2(1-s_{\mathrm{HGR}}), \\ m_1(A_3) &= 1-b_1; & m_2(A_3) &= 1-b_2, \\ m_1(A_4) &= 0; & m_2(A_4) &= 0, \end{aligned} \tag{13}$$

where b_1 and b_2 are hyperparameters to regulate the degree of confidence for the Euclidean score and HGRMC score. A higher m_1 or m_2 indicates a greater probability of the corresponding hypothesis.

Given the mass functions of the Euclidean and HGRMC scores, we employ the combination rule based on the DS evidence theory fusion rule [72] to obtain the final combined mass function as follows:

$$\begin{aligned} m_c(A_1) &= \frac{m_1(A_1)m_2(A_1)+m_1(A_1)m_2(A_3)+m_1(A_3)m_2(A_1)}{1-Z} \\ m_c(A_2) &= \frac{m_1(A_2)m_2(A_2)+m_1(A_2)m_2(A_3)+m_1(A_3)m_2(A_2)}{1-Z} \\ m_c(A_3) &= \frac{m_1(A_3)m_2(A_3)}{1-Z} \\ Z &= m_1(A_1)m_2(A_2)+m_1(A_2)m_2(A_1). \end{aligned} \tag{14}$$

In our method, the final retrieval result is determined by the plausibility function of the combined mass function, which is defined as

$$\begin{aligned} \mathrm{PLB}(A_1) &= \sum_{B\cap A_1\neq\emptyset} m_c(B) \\ &= m_c(A_1)+m_c(A_3), \end{aligned} \tag{15}$$

where a higher plausibility function value indicates a higher priority of the corresponding sample in the retrieval dataset. According to the ranking results of plausibility function values of samples in the retrieval dataset, we obtain the final retrieval results of the query image.

V. Performance Analysis of HGRMC and Euclidean Features

In this section, we analyze the comparative robustness of HGRMC and Euclidean distance under different data conditions to illustrate the necessity for feature fusion in enhancing retrieval robustness based on a synthetic dataset and the real dataset used in Section VI: CBRSIR_VS [20]. The use of synthetic data allows for controlled experimentations, providing a clear baseline for comparison. To generate synthetic data, we construct two classes of 10-D vectors, drawing from two Gaussian distributions. One class is centered around the mean vector of $[+1, +1, \ldots, +1]$, while the other is centered around the mean vector of $[-1, -1, \ldots, -1]$, each following a Gaussian distribution with the same standard deviation of 1. Then, we create the noisy dataset, where Gaussian perturbations with a power of 2 are introduced to half of the samples. Specifically, we randomly select 50% of the training data of each class for noise addition. This noisy dataset is then divided into training and test sets, with 1000 samples allocated for training and 100 samples reserved for testing. A three-layer linear neural network, which is composed of linear layers followed by rectified linear unit (ReLU) activation functions, is then trained on both the original and noisy datasets using either the Euclidean distance ranking loss or the HGRMC ranking loss. We use the Adam optimizer [78] with a learning rate of 0.001, a minibatch size of 256, and over a total of 100 epochs. After training, we visualize the distributions of learned features on the test dataset using the t-distributed stochastic neighbor embedding (T-SNE) method [79].

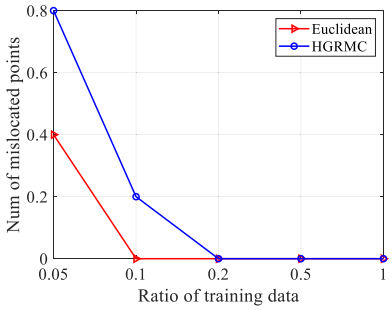


Fig. 4. Average number of mislocated points with different training data ratios.

Using the original synthetic dataset, we calculate the average number of mislocated data points in the test dataset of ten trials, which are not correctly grouped around the right cluster with different training data ratios. Mislocated data encompasses both misclassified points and outliers. A point is deemed misclassified if its average distance to members of the other class falls below that to members of its own class. Conversely, a point is labeled as an outlier if its smallest distance to any other dataset point surpasses the 75th percentile of all such distances. Each mislocated point, whether misclassified or an outlier, is counted only once in the total number of mislocated data points. As shown in Fig. 4, when the training data are rare, only using HGRMC can cause more mislocated points due to the difficulty of estimating mean and covariance matrices with little data provided than using Euclidean distance loss. Fig. 5 shows one typical T-SNE result when we use 10% of the full training dataset. Using Euclidean distance manages to better preserve the inherent structure of the data, whereas employing HGRMC features results in two outliers.

Fig. 6 shows the feature distribution with noisy synthetic datasets. The distribution derived from the Euclidean distance is visibly influenced by the noise, leading to several ambiguously placed data points. In contrast, the HGRMC approach largely retains distinct cluster formations with fewer mislocated points, underscoring the reliability of HGRMC in noisy environments.

In addition to the controlled experiments conducted with synthetic data, we extend our analysis to include real datasets used in Section VI. Fig. 7 presents the T-SNE results derived from employing only 1% of the real training dataset. As shown in Fig. 7, the T-SNE visualization generated by HGRMC reveals the formation of only five clusters from data comprising ten classes, while using Euclidean distance under the same condition yields improved category distinction, suggesting that the Euclidean distance is more adept at preserving inherent data structure when training data are scarce. On the other hand, Fig. 8 shows the robustness of the HGRMC approach

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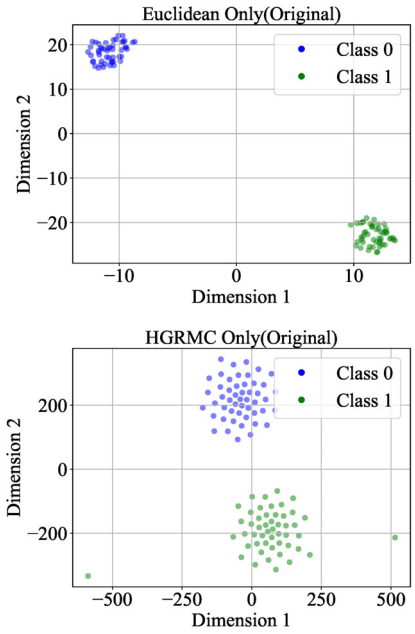


Fig. 5. T-SNE visualization of the feature distribution of the test dataset based on the model trained by 10% of the original generated training dataset.

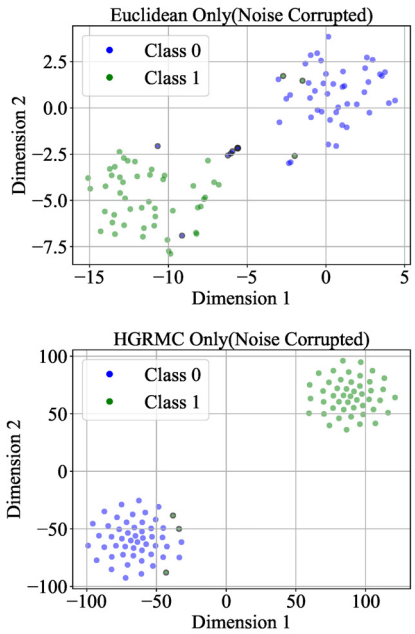


Fig. 6. T-SNE visualization of the feature distribution of the test dataset based on the model trained by the noisy generated training dataset.

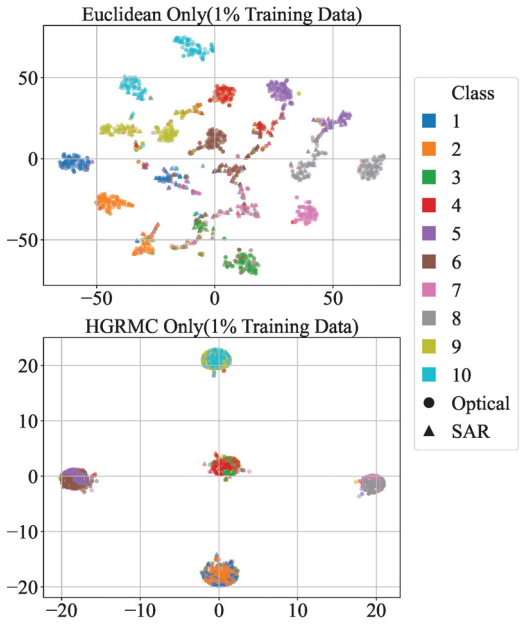


Fig. 7. T-SNE visualization of the feature distribution of the test dataset based on the model trained by 1% of the real training dataset used in Section VI.

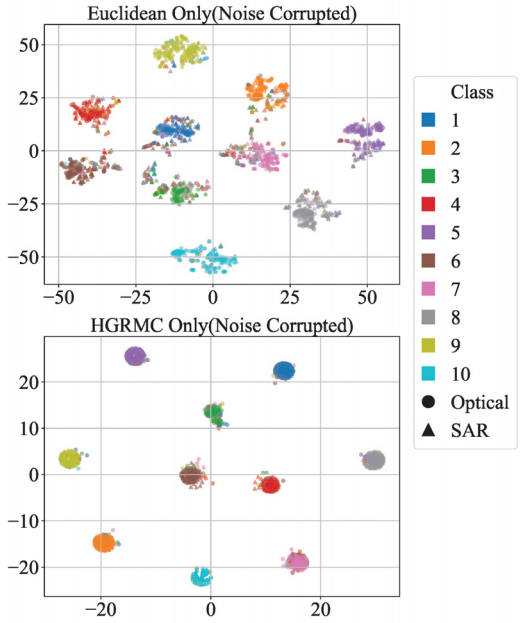


Fig. 8. T-SNE visualization of the feature distribution of the test dataset based on the model trained by the real noisy training dataset used in Section VI.

in noisy environments. Here, the HGRMC approach, when trained with the real noisy dataset, successfully delineates ten tight clusters, outperforming the result by using the Euclidean distance, which results in looser cluster formations. These conclusions are consistent with the results from the synthetic data experiments.

In summary, the HGRMC offers robustness in noisy environments, whereas the Euclidean distance offers stability when estimation from data is difficult. Combining these strengths ensures a consistent and reliable model performance.

VI. Experimental Results

A. Dataset

To validate the effectiveness of the proposed HAC method, we conduct experiments on a large dataset, CBRSIR_VS [20]. It consists of ten categories and a total of 26 901 pairs of optical and SAR images. Fig. 9 shows some examples in CBRSIR_VS [20]. The optical images are collected from Google Earth, while the SAR images are acquired from the Sentinel-1 satellite. Each optical image is with 256 × 256 pixels and 1-m resolution, and each SAR image is with 64 × 64 pixels and 10-m resolution. Both test and validation datasets contain randomly selected 1000 image pairs, where 100 image

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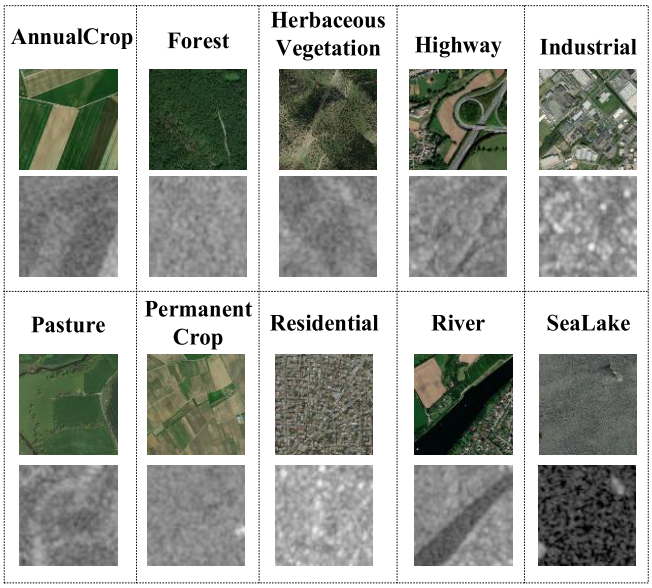


Fig. 9. Examples in CBRSIR_VS.

pairs are in each category. The remaining image pairs are allocated for the training purpose. We compare the proposed HAC method with ten state-of-the-art methods, including one traditional method: CCA [34], and nine DNN-based methods: deep CCA (DCCA) [36], correspondence autoencoder (Corr-AE) [37], adversarial cross-modal retrieval (ACMR) [42], dual modality collaborative learning (DMCL) [45], deep semantic matching (Deep-SM) [80], CMIR-Net [24], FGCrossNet [81], semantically supervised maximal correlation (S2MC) [69], and deep supervised cross-modal retrieval (DSCMR) [43]. CCA [34] is a classical unsupervised method that learns linear projections to maximize the correlation between two modalities. In contrast, the other methods are based on DNNs. Among DNN-based methods, DCCA [36] and Corr-AE [37] are unsupervised approaches that extend CCA [34] to learn nonlinear transformations of the input data. The remainder is supervised approaches that leverage label supervision to learn a common representation space for different modalities.

B. Implementation Details

We implement the experiments using the Pytorch framework on one NVIDIA A100. Each image is resized to a dimension of 224 × 224, ensuring compatibility with the input requirements of AlexNet. We extract 4096-D high-level feature vectors from the fc3 of the AlexNet model for each image. Then, two one-layer fully connected networks are used to generate the 512-D Euclidean features and the 512-D HGRMC features. The prediction layer is followed with another fully connected layer with ten output elements after the Euclidean features and HGRMC features. We use Adam [78] as the optimizer to train our retrieval model with the 512 batch size and 50 epochs. In addition, we applied a weight decay of 0.0001 for the optimizer to prevent overfitting of our model. The initial learning rate is set to 0.0001 for layers before fc3 of AlexNet and 0.0005 for the last two layers. A strategic linear decay of the learning rate commences at epoch 25, gradually reducing to zero by the end of training, facilitating a gradual reduction of the learning rate and smooth convergence. The hyperparameters of $\{\alpha,\ \beta,\ b_1,\ b_2\}$ are determined by the grid search strategy, where $\alpha = \beta = 0.001$ and $b_1 = b_2 = 0.95$ are used in the following experiments. To guarantee fairness in our comparisons, all the competing methods employ the same feature extraction module as the proposed method and we maintain identical training settings as those used for the proposed method. In addition, the size of output feature vectors for the competitive methods is standardized to $512 \times 2 = 1024$ to preserve fairness.

In order to evaluate the performance of our proposed retrieval method, we employ a set of widely used evaluation metrics, including the mean average precision (MAP), normalized discounted cumulative gain (NDCG), and the precision–recall (P–R) curve.

MAP is a metric for measuring the average retrieval accuracy across all queries. For each query, the average precision (AP) is calculated as

$$\mathrm{AP} = \frac{1}{K}\sum_{k=1}^{K} Pc(k)r(k), \tag{16}$$

where K is the number of retrieved images for a query sample. We consider all the samples in the test dataset when calculating AP in our experiments. $Pc(k)$ is the precision of the kth return sample (i.e., the ratio of relevant items among top-k returned samples), and $r(k)$ is an indicator function that equals 1 if the image at the rank k is relevant and 0 otherwise. MAP is the average AP value of all the queries in the test dataset.

NDCG measures the quality of the ranked retrieval results by considering both the relevance and the ranking positions. DCG of the kth return sample is calculated as

$$\mathrm{DCG}_k = \sum_{i=1}^{k} \frac{r(i)}{\log_2(i+1)}. \tag{17}$$

We calculate the DCG values for all the samples in the test dataset. We define the ideal DCG (IDCG) of the kth return sample as the maximum DCG value based on the ideal ranking. For one query, the NDCG of the kth return sample is the ratio of DCG and IDCG

$$\mathrm{NDCG}_k = \frac{\mathrm{DCG}_k}{\mathrm{IDCG}_k}. \tag{18}$$

We calculate the average NDCG value of all queries to evaluate different methods.

The P–R curve is a graphical representation that illustrates the tradeoff between precision and recall values using different thresholds. Precision is defined as the ratio of relevant retrieved items to the total number of retrieved items. Recall is the ratio of relevant retrieved items to the total number of relevant items in the dataset. We use the area under the P–R curve to evaluate the retrieval performance. An increased area corresponds to enhanced retrieval performance.

C. Comparisons With State-of-the-Art Methods

In this section, we present a comprehensive comparison between our proposed method and the state-of-the-art approaches using the CBRSIR_VS dataset. In order to ensure the reliability of our experimental results, we conduct three

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TABLE I
COMPARISON IN TERMS OF MAP AND NDCG SCORES

Method	MAP			NDCG		
	OPT to SAR	SAR to OPT	Average	OPT to SAR	SAR to OPT	Average
CCA [34]	0.1117	0.1066	0.1092	0.5938	0.5854	0.5896
DCCA [36]	0.1591	0.2313	0.1952	0.6373	0.6643	0.6508
Corr-AE [37]	0.2699	0.2651	0.2675	0.6782	0.6768	0.6775
ACMR [42]	0.8388	0.8141	0.8265	0.9556	0.9210	0.9383
DMCL [45]	0.8209	0.8365	0.8287	0.9445	0.9277	0.9361
Deep-SM [80]	0.8223	0.8425	0.8324	0.9477	0.9276	0.9377
FGCrossNet [81]	0.8431	0.8349	0.8390	0.9559	0.9259	0.9409
CMIR-Net [24]	0.8393	0.8439	0.8416	0.9526	0.9285	0.9406
S2MC [69]	0.8456	0.8432	0.8444	0.9558	0.9285	0.9421
DSCMR [43]	0.8432	0.8507	0.8470	0.9552	0.9299	0.9425
Our HAC	**0.8657**	**0.8661**	**0.8659**	**0.9637**	**0.9386**	**0.9512**

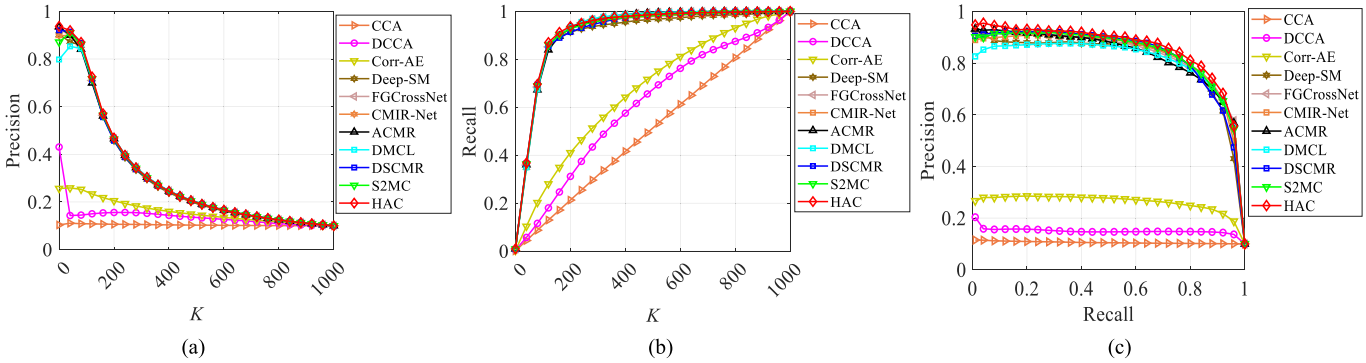


Fig. 10. Precision and recall curves for the optical-to-SAR image retrieval task. (a) Precision versus K. (b) Recall versus K. (c) P–R curves.

independent trials of our proposed method and the competitive state-of-the-art methods. The performance metrics, including MAP, NDCG, and P–R curves, are obtained for each trial. The final reported results represent the average performance across three trials.

Table I presents the MAP and NDCG scores obtained from experiments conducted in both optical-to-SAR and SAR-to-optical tasks, as well as the average performance of these two scenarios. CCA, which relies on single-layer models, struggles in complex cases due to its inability to capture high-level semantic information. DCCA and Corr-AE that employ DNN-based features in an unsupervised manner show improved retrieval performance than CCA but still lack a good capability to capture the relationship between modalities, as reflected in its lower MAP and NDCG in Table I. By integrating supervised information, DSCMR jointly learns feature representations to construct a common space of CM data and further improves the retrieval performance than unsupervised methods, as demonstrated by its superior improvements both in MAP and NDCG than unsupervised methods. Table I shows that the MAP scores achieved by our HAC method outperform those of the competitive approaches on both the optical-to-SAR and SAR-to-optical tasks. Our proposed method achieves an average improvement of 1.89% in terms of MAP compared with the other methods. Moreover, our method yields better NDCG values than other methods, which means that the proposed method has superior performance in both retrieving relevant images accurately and rank retrieved results appropriately.

Figs. 10 and 11 show the precision and recall values with a different number of returned samples K and P–R curves of different methods for both the optical-to-SAR and SAR-to-optical retrieval tasks. Our proposed HAC method consistently achieves higher precision and recall at various ranking positions compared with the competitive methods. Furthermore, the P–R curves of our proposed method are higher than those of other methods at different recall levels.

In Fig. 12, we present a visual comparison of the top ten retrieval results of four query samples obtained by our proposed HAC method and another two state-of-the-art competing methods, DSCMR [43] and S2MC [69]. As shown in Fig. 12, the samples retrieved by the proposed method are semantically accurate and similar to the query images in both optical-to-SAR and SAR-to-optical tasks, while the competitive methods

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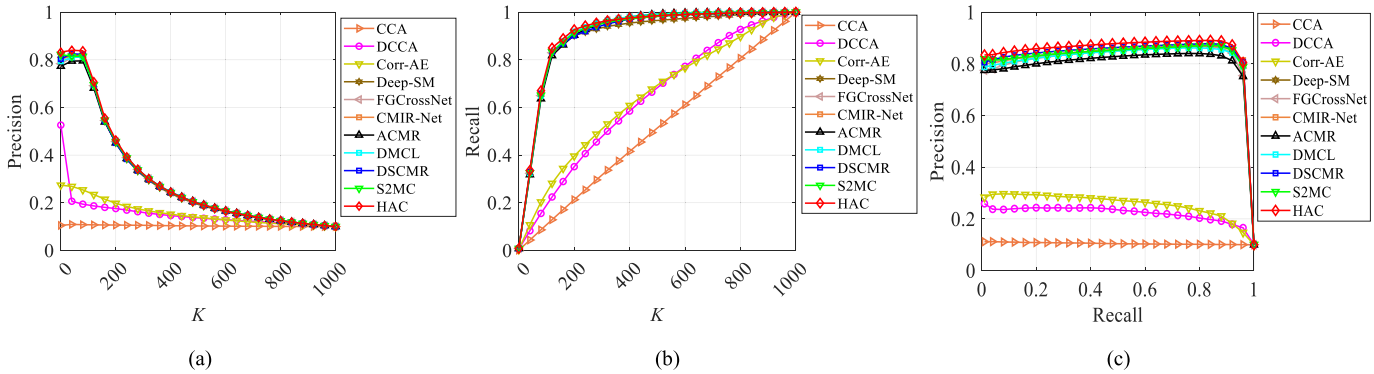


Fig. 11. Precision and recall curves for the SAR-to-optical image retrieval task. (a) Precision versus K. (b) Recall versus K. (c) P–R curves.

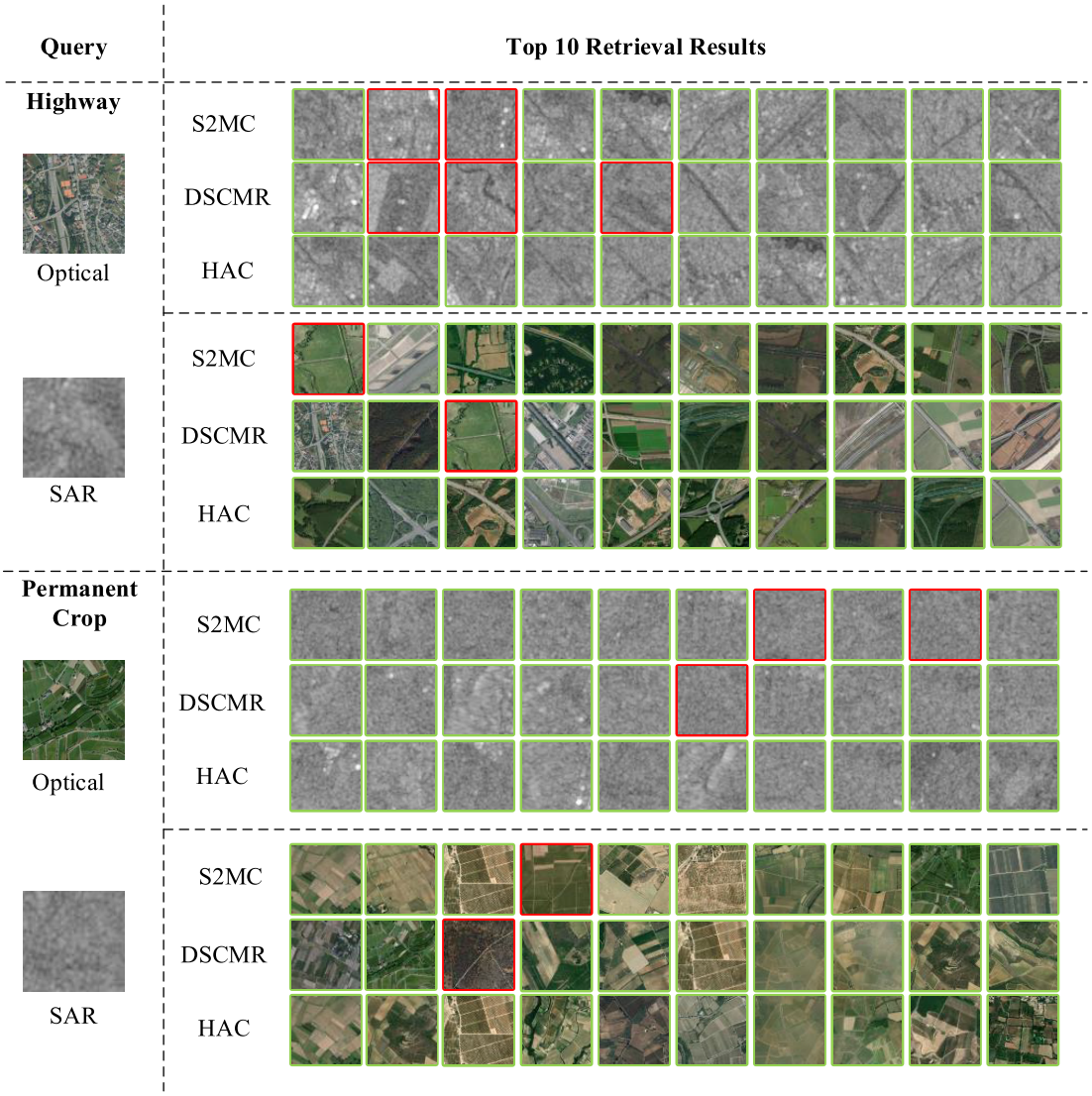


Fig. 12. Query examples with the corresponding top-10 retrieved images, where the samples in green rectangles are correct returned images, and samples in red rectangles are false positives.

also retrieve most of relevant samples, but with some false samples.

D. Ablation Study

In order to investigate the impact of different parts of the proposed method on retrieval performance, we perform the ablation experiments by evaluating the retrieval results after removing specific model components. As shown in Table II, when we remove the prediction loss, there is a significant drop in retrieval performance, demonstrating the importance of prediction loss in learning modality-independent feature representations and guaranteeing semantic consistency across

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TABLE II
RETRIEVAL PERFORMANCE WITH DIFFERENT COMBINATIONS OF LOSS TERMS, WHERE "W/O" MEANS "WITHOUT"

Loss terms	MAP			NDCG		
	OPT to SAR	SAR to OPT	Average	OPT to SAR	SAR to OPT	Average
w/o $\mathcal{J}_{HGR}$	0.8477	0.8500	0.8489	0.9551	0.9283	0.9417
w/o $\mathcal{J}_{EUC}$	0.8634	0.8538	0.8586	0.9611	0.9322	0.9467
w/o $\mathcal{J}_{P}$	0.3827	0.3976	0.3902	0.7586	0.7458	0.7522
w/o $\mathcal{J}_{R}$	0.8587	0.8599	0.8593	0.9558	0.9346	0.9452
w/o $\mathcal{J}_{CR}$	0.8502	0.8576	0.8539	0.9551	0.9336	0.9444
w/o $\mathcal{J}_{I}$	0.8527	0.8566	0.8547	0.9580	0.9348	0.9464
w/o $\mathcal{J}_{C}$	0.8492	0.8578	0.8535	0.9560	0.9349	0.9455
w/o $\mathcal{J}_{HGR}$, $\mathcal{J}_{EUC}$	0.8000	0.8420	0.8210	0.9391	0.9265	0.9328
w/o $\mathcal{J}_{R}$, $\mathcal{J}_{CR}$	0.8206	0.8447	0.8327	0.9453	0.9349	0.9401
w/o $\mathcal{J}_{I}$, $\mathcal{J}_{C}$	0.8048	0.8269	0.8159	0.9354	0.9197	0.9276
Full HAC	**0.8657**	**0.8661**	**0.8659**	**0.9637**	**0.9386**	**0.9512**

TABLE III
RETRIEVAL PERFORMANCE VERSUS FUSION STRATEGIES OF MAP AND NDCG SCORES

Method	MAP			NDCG		
	OPT to SAR	SAR to OPT	Average	OPT to SAR	SAR to OPT	Average
Only f_E , g_E	0.8554	0.8641	0.8598	0.9580	0.9373	0.9477
Only f_H , g_H	0.8617	0.8603	0.8610	0.9623	0.9363	0.9493
HAC sum	0.8624	0.8626	0.8625	0.9626	0.9373	0.9500
HAC DS	**0.8657**	**0.8661**	**0.8659**	**0.9637**	**0.9386**	**0.9512**

modalities. The introduction of similarity loss $\mathcal{J}_I$ results in a more significant increase in MAP compared to the introduction of the ranking loss $\mathcal{J}_R$. Similarly, introducing correlation loss $\mathcal{J}_C$ leads to a greater improvement in retrieval accuracy than introducing correlation ranking loss $\mathcal{J}_{\mathrm{CR}}$. This highlights the value of learning the knowledge from paired data from different modalities. The retrieval performance experiences a more pronounced improvement when the HGRMC loss is introduced into the model compared with the case when the Euclidean loss is introduced. This suggests that the HGRMC loss plays a more critical role in enhancing the retrieval performance.

We examine the retrieval performance using different combinations of features and fusion schemes in the retrieval phase, as presented in Table III. We compare the use of only Euclidean features, only HGRMC features, direct summation of the Euclidean score and the HGRMC score (HAC sum), and the proposed DS evidence theory-based fusion (HAC DS) in terms of MAP and NDCG scores. Table III indicates that the DS evidence theory-based fusion achieves the better performance in terms of both MAP and NDCG scores in comparison with direct summation of two scores, proving its effectiveness in fusing scores from different types of information. In addition, using only Euclidean or HGRMC features results in suboptimal retrieval performance compared to the fusion techniques.

We evaluate the impact of hyperparameters $\{\alpha, \beta, b_1, b_2\}$ on the retrieval performance of our proposed HAC method. The parameters α and β are introduced to balance the effect of different loss terms. We conduct experiments by varying the values of α and β from 0.0001 to 1 with an interval of ten times the previous value. When evaluating one hyperparameter, we keep others constant at the optimal values in our experiment (e.g., $\alpha = \beta = 0.01$ and $b_1 = b_2 = 0.95$). As shown in Fig. 13, the optimal retrieval performance is achieved when α and β are set with equal proportions, maintaining a balanced configuration. We also analyze the impact of hyperparameters b_1 and b_2 on the retrieval performance

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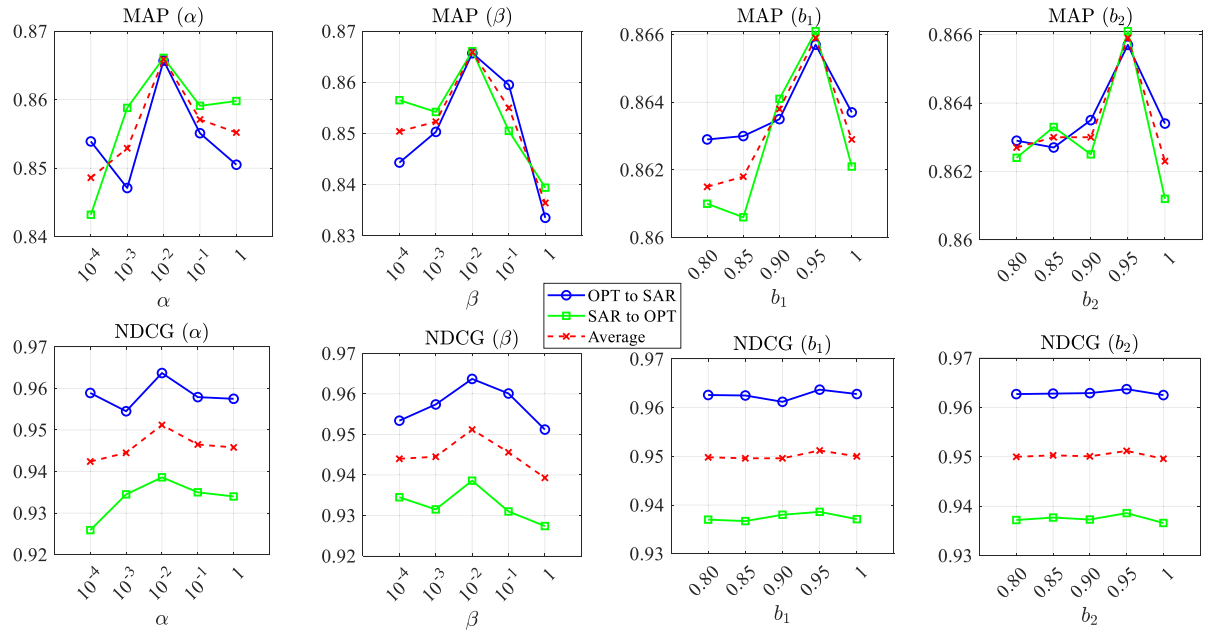


Fig. 13. Retrieval performance of parameter analysis in terms of MAP and NDCG scores.

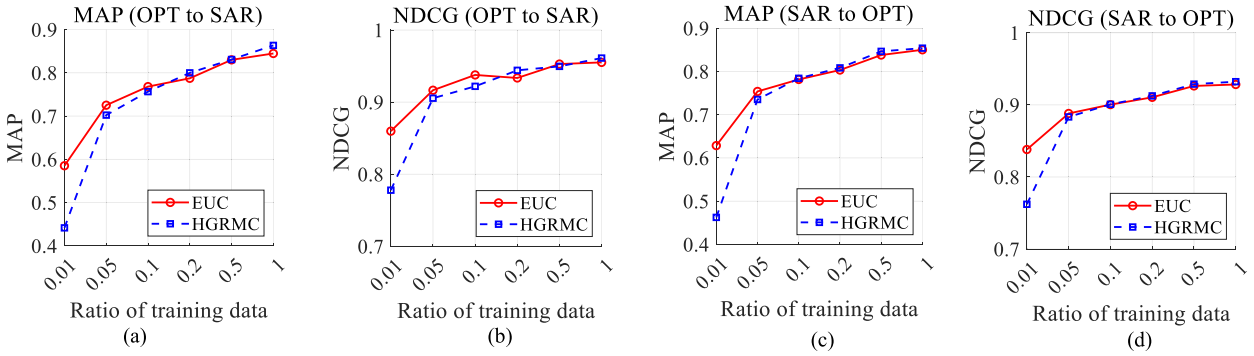


Fig. 14. Comparative retrieval performance of HGRMC and Euclidean features with different training data ratios. (a) MAP: optical-to-SAR. (b) NDCG: optical-to-SAR. (c) MAP: SAR-to-optical. (d) NDCG: SAR-to-optical.

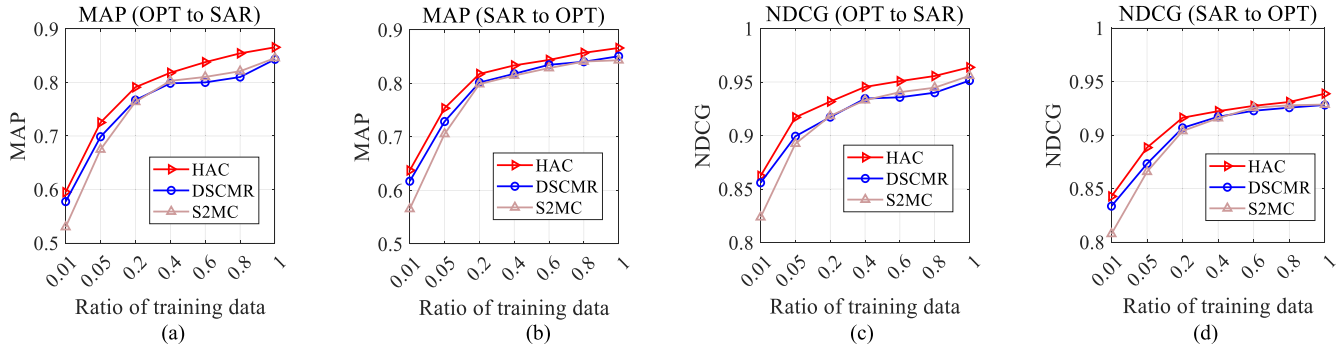


Fig. 15. Retrieval performance with different training data ratios. (a) MAP: optical-to-SAR. (b) MAP: SAR-to-optical. (c) NDCG: optical-to-SAR. (d) NDCG: SAR-to-optical.

of our proposed HAC method. The parameters b_1 and b_2 are used to regulate the degree of confidence for the Euclidean and HGRMC scores, respectively. We conduct experiments by varying the values of b_1 and b_2 from 0.8 to 1 with an interval of 0.05. As shown in Fig. 13, the retrieval performance is not significantly influenced by $b_1, b_2 \in [0.8, 1]$, indicating the robustness of our method to these parameters. The optimal retrieval performance is achieved when both b_1 and b_2 are set to 0.95.

To evaluate the robustness of feature learning under different constraints and data provisions, we compare the retrieval performance using varying training dataset sizes, relying solely on either HGRMC or Euclidean features. Fig. 14 illustrates that, especially when data are limited (using only 1% of

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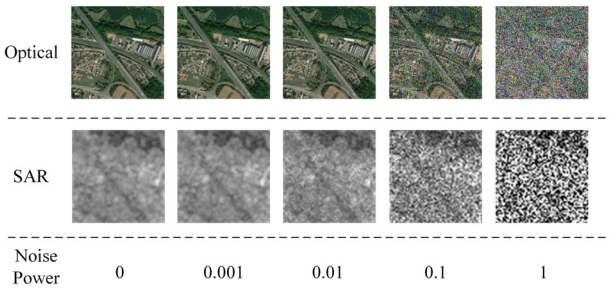


Fig. 16. Data examples with different noise intensity of CBRSIR_VS.

the original training dataset), retrieval based on Euclidean features surpasses that of HGRMC in terms of MAP and NDCG. In such scenarios with rare training data, the MAP for Euclidean features can be approximately 15% higher than that for HGRMC. This disparity arises because accurately estimating means and covariance from limited data is challenging for HGRMC, while Euclidean distance does not rely on statistical estimation. As the amount of training data increases, the performance gap diminishes. With a more substantial data provision, HGRMC either matches or slightly outperforms the Euclidean features. These experimental findings align with the results presented in Section V.

E. Effect of Training Data Ratio Analysis

In order to analyze the effect of training data ratios on the retrieval performance of our proposed method, we train our model with different ratios of training data. We compare the retrieval results of our HAC with the second-best method DSCMR [43] and the third-best method S2MC [69] in Table I in terms of MAP and NDCG. The ratios of training data for investigation are 1%, 5%, 20%, 40%, 60%, 80%, and 100%, and we keep validation and test data unchanged. As shown in Fig. 15, our proposed method outperforms the other two competitors for different training data ratios. This demonstrates that our method better utilizes the available training data to enhance the retrieval performance. The retrieval accuracy of S2MC decreases more significantly than the other two methods because it relies solely on a correlation metric that requires the estimation of statistics, which becomes less reliable with scarce data. In contrast, our proposed method can still achieve better retrieval performance with fewer training data, demonstrating its robustness and stability to handle scenarios with limited data availability.

F. Noisy Scenario

In order to evaluate the robustness of our proposed HAC method against low-quality images with strong noise, we conduct experiments under various noise conditions. We introduce Gaussian noise to optical images and speckle noise to SAR images, each with a variance that systematically increases from 0.001 to 1, at increments multiplying the variance tenfold at each step. This approach is designed to replicate typical noise patterns encountered in real-world applications [47]. After the noise is introduced, we normalize the images again to ensure that the pixel values remain within the range [0, 1]. The noise introduction is conducted on both training and test

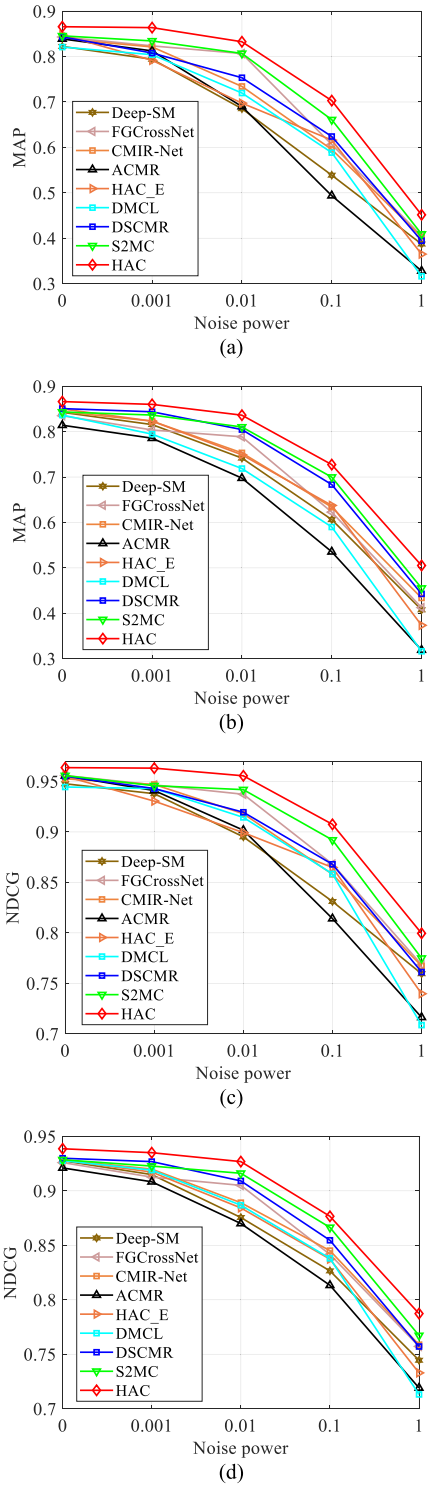


Fig. 17. Retrieval performance with different noise power. (a) MAP: optical–to-SAR. (b) MAP: SAR-to-optical. (c) NDCG: optical-to-SAR. (d) NDCG: SAR-to-optical.

datasets, ensuring that our model's performance is evaluated under consistent noise-affected conditions. Fig. 16 presents the examples of data with different noise intensities, where the noise significantly affects the quality of the images as the intensity increases, making the retrieval task more challenging.

In Fig. 17, we compare our method with the seven best-performing methods from our previous experiments to assess the retrieval performance under noisy conditions. We also

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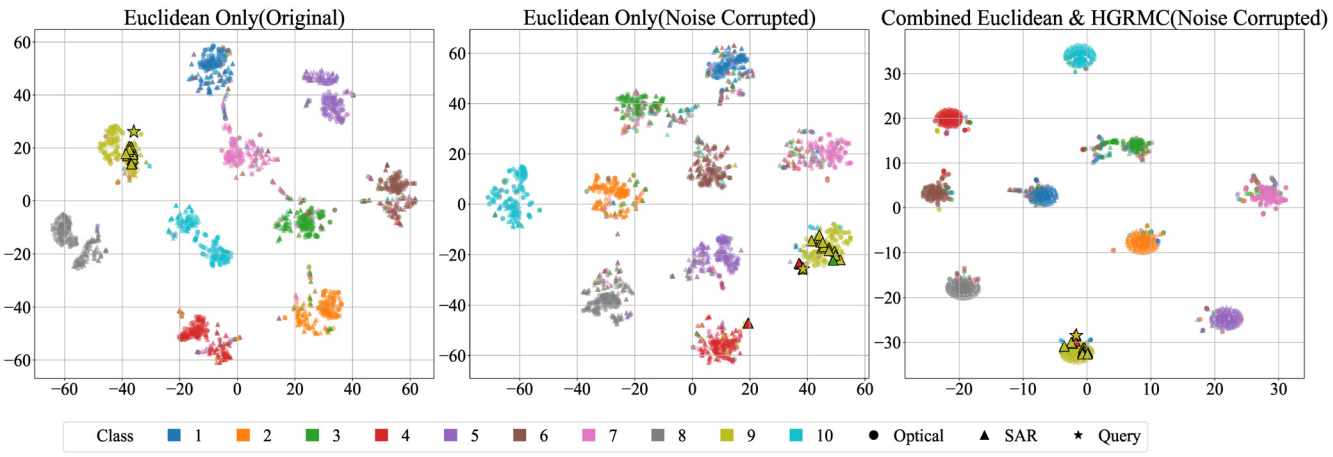


Fig. 18. T-SNE visualization of features based on Euclidean distance and a combination of Euclidean and HGRMC features under original and noisy scenarios. The optical modality is represented by circles, while SAR is represented by triangles. An example query image from the optical modality is denoted by a star, with its top-20 retrieval results in the SAR modality highlighted.

compare the proposed method without the HGRMC loss to demonstrate the role of HGRMC loss in enhancing the noise resistance, named HAC_E. From Fig. 17, it is evident that our method consistently achieves higher MAP and NDCG scores under noisy conditions compared with other methods. For the optical-to-SAR retrieval task, when we do not introduce extra noise, our method achieves an MAP score of 0.8657, approximately 2.4% higher than that of the second-best method, S2MC, which scores 0.8456. As the noise intensity increases, the performance gap widens significantly, underscoring the pronounced advantages of our proposed model in handling noisy conditions. A similar trend is observed in the SAR-to-optical retrieval task. As shown in Fig. 17(c) and (d), the NDCG scores also exhibit a similar trend as the MAP scores and demonstrate the superiority of our proposed method in terms of retrieval robustness again. In addition, the results clearly show that the retrieval performance of HAC_E significantly decreases in the absence of the HGRMC loss, demonstrating the critical role of HGRMC loss in enhancing the noise resistance of our proposed model.

To demonstrate the resilience of the proposed method against noise, we perform a T-SNE visualization of the Euclidean and HGRMC features under both original and noisy scenarios with a noise power of 0.1, as shown in Fig. 18. The data points from the optical and SAR modalities are represented by circles and triangles, respectively. We choose a typical query image from the optical modality, marked with a star, and its top-20 retrieved results in the SAR modality are also highlighted for clarity. In the scenario without extra noise, using the Euclidean features correctly retrieves the associated samples. However, when noise is introduced, the Euclidean features demonstrate a dispersion of points, with some even offset to the clusters of other classes. This suggests that noise introduces disturbances in Euclidean feature learning, leading to a decrease in retrieval accuracy. Conversely, using the fusion of the HGRMC features and Euclidean features shows remarkable resilience against noise. The top-20 retrieval results for the chosen query image, despite the presence of noise, remain tightly clustered, indicating the correct retrieval. This demonstrates that the integration of the HGRMC offers enhanced robustness in the presence of noise.

VII. Conclusion

In this article, we proposed a novel HGRMC augmented method for the CM-CBRSIR task. Most previous studies only optimize features for CM-CBRSIR based on Euclidean distances appreciated for its computational efficiency and simplicity without the need for statistical estimations. Despite these strengths, relying solely on Euclidean distances can lead to nonrobust retrieval performance, particularly in noisy scenarios. We addressed this issue by not only extracting the Euclidean distance-based features but also the HGRMC-based features to enhance robustness in noisy scenarios. The newly considered HGRMC-based features emphasize maximizing the correlation and resist noise distortions through normalization and orthogonal learning, effectively capturing information from distinct modalities. In the retrieval phase, Euclidean distance and HGRMC-based retrieval scores are fused based on the DS evidence theory. Extensive experiments on a large-scale CM-CBRSIR dataset show that our proposed HAC algorithm outperformed existing state-of-the-art CM-CBRSIR methods, especially in the scenarios with low-quality noisy images. Future efforts will focus on adaptive feature fusion to improve retrieval accuracy and efficiency, advancing from our current methodology of manual grid search for the selection of fusion parameters in the retrieval phase.

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